

EventSmartKafka

A Custom Kafka Consumer Adapter for SAP BTP Integration Suite

Enterprise Kafka Connectivity for SAP Cloud Integration

EXECUTIVE SUMMARY

- 1.5M+ messages validated • Multiple Kafka security profiles
- SAP Cloud Connector TCP/SOCKS5 • Avro Conversion
- Cloud and On-Premise Scenarios

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The Problem

Enterprise Kafka Connectivity
from SAP CPI

Enterprise Kafka Connectivity from SAP CPI



Cloud vs On-Premise

Bridging cloud-native SAP CPI with on-premise Kafka clusters introduces network, routing and latency challenges



Security Complexity

SASL, TLS, SCRAM-SHA-256/512 — each adds configuration surface that standard adapters rarely expose



Schema Registry

Avro deserialization with Confluent Wire Format and Schema Registry adds a governance layer most adapters skip



Operational Visibility

Without native Kafka consumer support, CPI lacks partition-level metrics, offset tracking and message-level diagnostics



Native TCP Requirements

SAP ADK provides HTTP proxying. Kafka uses raw TCP binary protocol. This protocol gap is the fundamental integration challenge



Scale Challenges

Traditional HTTP-based workarounds break down under load: timeouts, connection leaks, and opaque error handling

The core challenge: SAP CPI needs a native Kafka consumer that speaks TCP, handles enterprise security, understands Avro schemas, and provides operational visibility — none of which exist in the standard adapter portfolio.

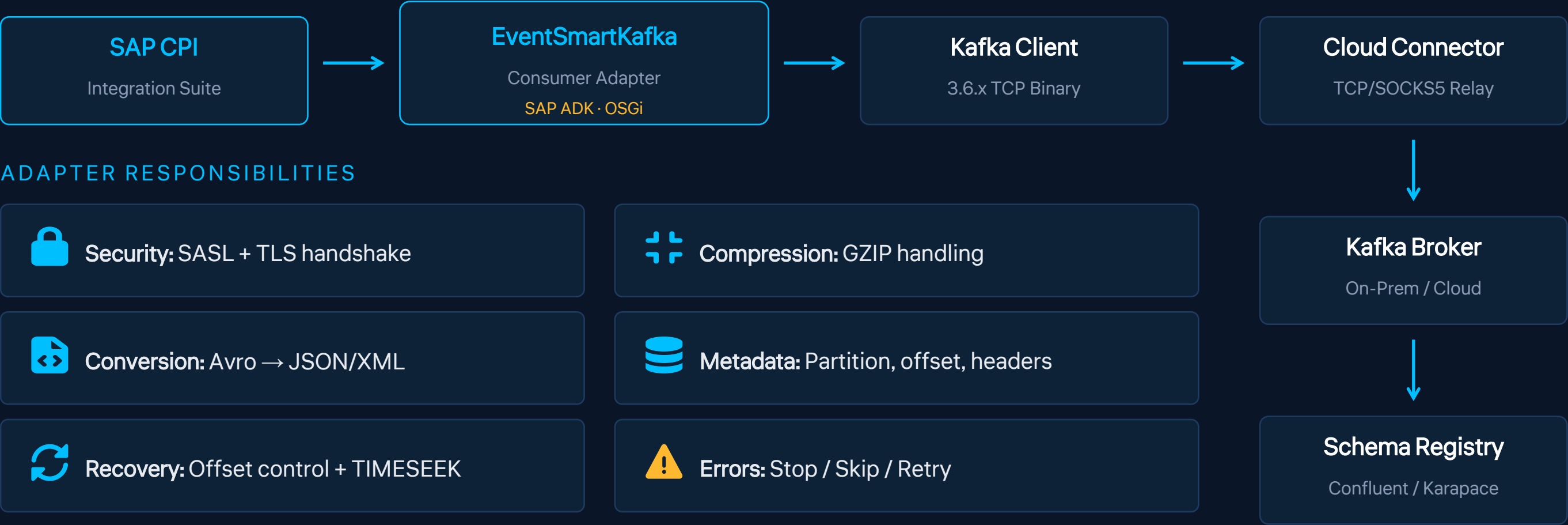
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The Architecture

End-to-End Integration Flow

Pipeline Design

End-to-End Integration Flow



The adapter acts as a **protocol bridge** — translating SAP CPI's integration model into Kafka's native TCP wire format, while managing security, schema conversion, and operational state.

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SDIA · EDCP · ODCP

Governance Model

Frameworks & Patterns

SDIA · EDCP · ODCP

FRAMEWORK

SDIA

Semantic Domain Integration Architecture

The overarching governance framework that defines how integration domains are structured and governed

PATTERN

EDCP

Event Domain-Centric Pattern

How event-driven integrations are organized around domain boundaries — EventSmartKafka implements this pattern

PATTERN

ODCP

Orchestration Domain-Centric Pattern

How orchestrated workflows interact with domain-centric event flows — the complementary pattern to EDCP

CORE PRINCIPLE

One Domain · One Grammar · Multiple Artifacts

HOW EVENTSMARTKAFKA FITS



Domain: Kafka Event Ingestion — a single bounded integration domain



Grammar: TCP/SASL/Avro/Offset — the protocol language of the domain



Artifacts:

Adapter · iFlow · Security Profile · Schema · Topic · Offset Policy

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Key Capabilities

Security · Connectivity · Conversion

Operational Control



CAPABILITIES

Key Capabilities at a Glance

AUTHENTICATION

PLAINTEXT

SASL_PLAINTEXT

SASL_SSL

PLAIN

SCRAM-SHA-256

SCRAM-SHA-512

CONNECTIVITY & CONVERSION



Cloud Connector
TCP/SOCKS5 relay for on-premise



Schema Registry
Confluent / Karapace support



Avro Conversion
Avro → JSON and Avro → XML

OPERATIONAL CONTROL



Offset Control
Commit after successful processing



TIMESEEK Recovery
Seek by timestamp for replay



Error Handling
Stop / Skip / Retry modes



TLS Trust Modes: JVM Default Truststore · Custom CA (PEM) · CPI Keystore Alias · Separate Schema Registry credentials over TLS



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Test Environment

Validation Setup

Infrastructure & Configuration

Test Environment Setup

INFRASTRUCTURE

 Local Kafka

 Docker

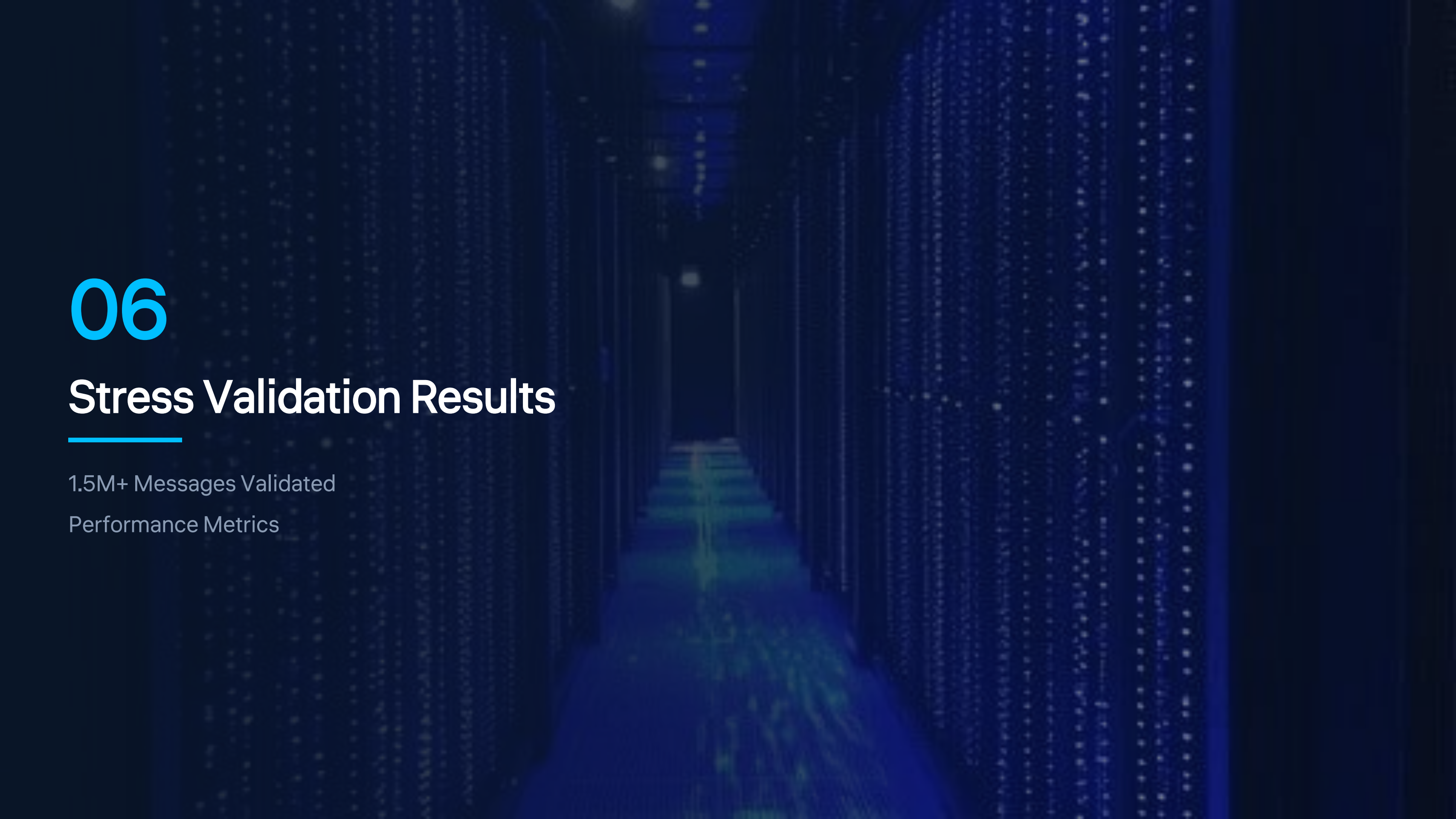
 SAP Cloud Connector

 SAP BTP Trial

TEST CONFIGURATIONS

Dimension	Scope	Details
CPI iFlows	Multiple integration flows	Tested across different iFlow designs and patterns
Security Profiles	Multiple auth configurations	PLAINTEXT, SASL_PLAINTEXT, SASL_SSL with SCRAM
Topics	Multiple Kafka topics	Different partitioning and replication strategies
Avro Payloads	53 KB · 1 MB · 2 MB	Small, medium, and large message sizes
Schema Registry	Confluent-compatible	Magic Byte and Fixed Schema ID resolution
Environment	Controlled stress test	Monitored JVM, OSGi, and CPI runtime metrics

Controlled Stress Environment — All tests executed with runtime monitoring: JVM heap, GC cycles, OSGi bundle state, CPI message queue depth, and Kafka consumer lag metrics



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Stress Validation Results

1.5M+ Messages Validated

Performance Metrics

RESULTS

Stress Validation Results

1.5M+

Messages Validated

Multiple

Tunnels & iFlows

3 Sizes

Payload Variants Tested

2 Modes

Cloud & On-Premise

VALIDATION METRICS

Metric	Produced	Consumed	Observation
Small (53 KB)	500K+	500K+	Zero loss, stable throughput
Medium (1 MB)	600K+	600K+	Consistent offset commit behavior
Large (2 MB)	400K+	400K+	GZIP + Schema Registry stable
Failed Messages	< 0.01%	—	Error handling (Skip/Retry) effective
Retries	< 0.05%	—	Transient failures recovered
Runtime Stability	Validated	Validated	OSGi + CPI runtime stable

Key Finding: The adapter maintained stable memory consumption, consistent offset commits, and zero OSGi bundle resolution errors across all 1.5M+ messages.



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Operational Lessons Learned

Real-World Findings

Troubleshooting & Tuning

Lessons Learned

01 Advertised Listeners

Kafka broker listener config must match Cloud Connector tunnel endpoints

02 Cloud Connector TCP Relay

SOCKS5 proxy behavior affects connection establishment timing

03 Metadata Timeout Troubleshooting

Default metadata timeout too aggressive for Cloud Connector latency

04 Offset Commit Tuning

Commit-after-process pattern prevents data loss but needs tuning

05 Schema Registry Tuning

GZIP decompression and caching reduce Schema Registry round-trips

06 Large Payload Handling

2 MB payloads require adjusted fetch size and memory buffers

07 OSGi Runtime Lifecycle

Bundle start/stop, refresh, and CPI deployment cycles require careful resource cleanup to prevent Kafka consumer thread leaks



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What Makes This Different

Differentiators & Value Proposition

Enterprise Integration

What Makes This Different



Built Specifically for SAP CPI

Not a generic Kafka client — designed for SAP BTP Integration Suite



Native Kafka TCP Support

Speaks raw TCP binary protocol — not HTTP workarounds



Cloud Connector Integration

First-class TCP/SOCKS5 relay support for on-premise brokers



Production-Oriented Troubleshooting

Designed from real operational pain points, not demo scenarios



Open Source

Apache License 2.0 — inspect, modify, contribute



Governance Aligned to SDIA

Follows Semantic Domain Integration Architecture principles



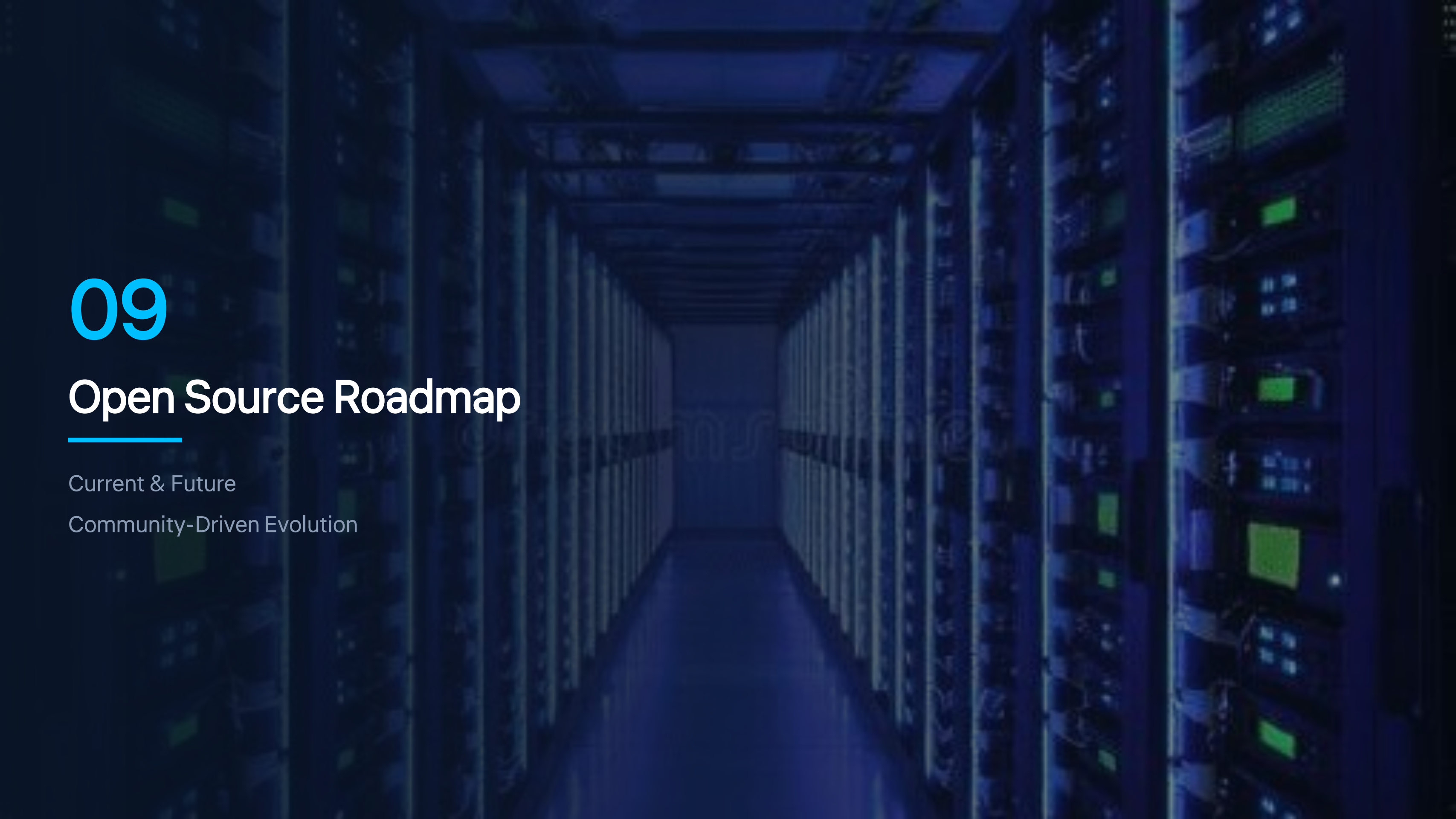
Operational Observability

Compact diagnostics — structured metadata, not verbose dumps



Community-Driven Evolution

Grows with community feedback and real-world usage



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Open Source Roadmap

Current & Future

Community-Driven Evolution

Open Source Roadmap

CURRENT — VALIDATED

 **Kafka**

 **Avro**

 **Schema Registry**

 **Cloud Connector**

FUTURE — PLANNED

**Additional Event Brokers**
Beyond Kafka — extensible adapter pattern

**Additional Payload Formats**
Protobuf, JSON Schema, custom formats

**Advanced Monitoring**
Metrics, dashboards, alerting hooks

**Extended Governance**
Deeper SDIA/EDCP/ODCP integration

**Community Contributions**
Open PRs, issues, and feature requests

Community Invitation: This roadmap is shaped by real-world needs. Contributions, feedback, and production experiences from the SAP SCN Community drive prioritization.

One Domain. One Grammar. Multiple Artifacts.

EventSmartKafka demonstrates how enterprise event-driven integration
can combine **governance, observability, connectivity**
and **operational simplicity** inside SAP BTP Integration Suite.

FRAMEWORKS & PATTERNS

SDIA — Semantic Domain Integration Architecture

EDCP — Event

ODCP — Orchestration

DEIP — Domain Enterprise Integration Pattern

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Thank You